Machine Learning Model Compression and Optimization for Vehicle Fuel Efficiency Prediction

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Abstract

Fuel efficiency prediction is an important application of machine learning in transportation, environmental sustainability, and automotive engineering. Accurate prediction models can assist manufacturers in improving vehicle designs and help consumers make informed purchasing decisions. This project investigates machine learning and neural network approaches for vehicle fuel efficiency prediction using two datasets: the Auto MPG dataset and the Vehicles Fuel Efficiency dataset. Multiple models, including Linear Regression, Support Vector Regression (SVR), Random Forest Regression, and Multilayer Perceptron (MLP), were implemented and evaluated. Compression and optimization techniques such as Principal Component Analysis (PCA), coreset selection, pruning, and dimensionality reduction were explored to reduce model complexity while preserving predictive performance. Furthermore, a PCA-KNN pipeline was proposed as an improvement for the Vehicles dataset. Experimental results indicate that Random Forest consistently achieved the highest predictive accuracy across both datasets, while PCA significantly enhanced neural network performance on high-dimensional vehicle data.

1 Introduction

Fuel efficiency has become an increasingly important concern due to rising fuel costs, environmental regulations, and the growing need for sustainable transportation solutions. Predicting fuel efficiency accurately enables manufacturers to optimise vehicle design and provides consumers with valuable information during purchasing decisions.

Machine learning techniques have shown significant potential in modelling the complex relationships between vehicle characteristics and fuel consumption. Traditional statistical approaches often struggle to capture nonlinear interactions present in automotive datasets, whereas machine learning methods can learn these relationships effectively.

This project investigates and compares several machine learning and neural network techniques for predicting vehicle fuel efficiency. Beyond prediction accuracy, the study also explores compression and optimisation methods aimed at reducing computational complexity while maintaining acceptable performance levels.

The objectives of this study were:

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- To compare the predictive performance of several machine learning models.
- To investigate the effectiveness of neural network compression techniques.
- To evaluate the impact of dimensionality reduction on high-dimensional datasets.
- To propose and assess an improved modelling pipeline for vehicle fuel efficiency prediction.

The major contributions of this project include:
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- Comparative evaluation of multiple machine learning models on two distinct datasets.
- Investigation of PCA, coreset selection, and pruning techniques.
- Development of a PCA-KNN pipeline for high-dimensional vehicle data.
- Comprehensive analysis of how dataset characteristics influence model performance.

2 Literature Review

Machine learning has been widely applied to fuel efficiency prediction problems.

2.1 Aliyu, Adeshina and Siraj (2014)

Aliyu et al. investigated the use of neural networks for fuel efficiency prediction using the Auto MPG dataset. Their study approached the problem as a classification task by grouping vehicles into efficiency categories. A Multi-Layer Perceptron trained using backpropagation demonstrated strong classification capability. The authors also reported that removing outliers improved model generalisation.

2.2 Sarkar and Hasan (2025)

Sarkar and Hasan compared multiple machine learning algorithms using both the classic Auto MPG dataset and more modern vehicle datasets. Their findings highlighted that vehicle characteristics remain strong predictors of fuel efficiency despite advances in automotive technology. Ensemble methods were found to perform particularly well.

2.3 Çeken and Gemci (2025)

Çeken and Gemci combined the Auto MPG dataset with EPA Fuel Economy data containing over 38,000 vehicle records. The authors evaluated several machine learning and deep learning approaches including Random Forest, Artificial Neural Networks, CNNs, and XGBoost. XGBoost achieved the strongest regression performance, while CNN performed well in classification tasks.

2.4 Research Gaps

Although previous studies demonstrated promising predictive capabilities, several limitations remain:
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- Limited inclusion of modern hybrid and electric vehicles.
- Insufficient investigation of model compression techniques.
- Limited exploration of dimensionality reduction strategies.
- Lack of focus on balancing predictive accuracy with computational efficiency.

This study addresses these gaps by evaluating compression techniques and proposing a PCA-KNN improvement strategy.

3 Datasets

3.1 Auto MPG Dataset

The Auto MPG dataset contains 398 vehicle observations collected during the 1970s and early 1980s. The target variable is Miles Per Gallon (MPG). Features include cylinders, displacement, horsepower, weight, acceleration, model year, and origin.

The dataset contains only six missing horsepower values and represents a relatively simple, low-dimensional regression problem.

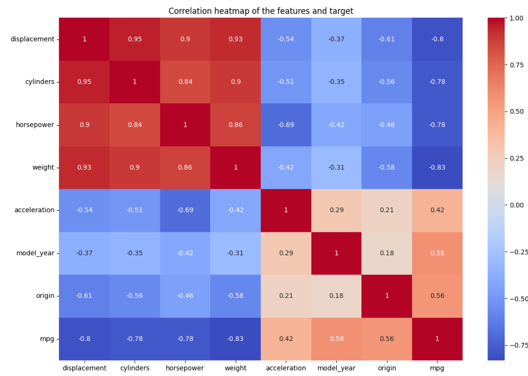

Figure 1:

Table 1: Auto MPG Dataset Summary

Property	Value
Observations	398
Attributes	9
Target variable	MPG
Missing values	Minimal

3.2 Vehicles Fuel Efficiency Dataset

The Vehicles dataset contains 47,702 observations and 84 attributes. The target variable used in this study was Combined Fuel Efficiency (`comb08`). The dataset includes numerical, categorical, and Boolean variables, as well as substantial missing values. Preprocessing procedures included missing value handling, label encoding, feature scaling, and dimensionality reduction.

The complexity of this dataset provided an opportunity to investigate how machine learning models behave under high-dimensional conditions.

4 Methodology

4.1 Machine Learning Models

Linear Regression. A baseline model used to establish the performance of a simple linear approach.

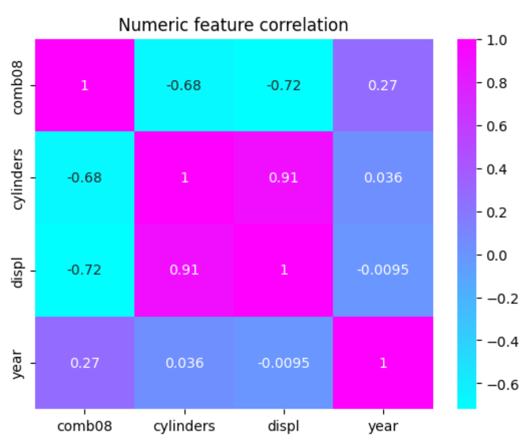

Figure 2:

Table 2: Vehicles Fuel Efficiency Dataset Summary

Property	Value
Observations	47,702
Attributes	84
Target variable	`comb08`
Missing values	Extensive

Support Vector Regression (SVR). SVR was used to capture nonlinear relationships through kernel transformations.

Random Forest Regression. An ensemble learning approach combining multiple decision trees to improve predictive accuracy and reduce overfitting.

4.2 Neural Network Models

Standard MLP. The baseline neural network consisted of fully connected hidden layers trained using the Adam optimiser.

PCA-Based MLP. PCA was applied before training to reduce dimensionality.

Coreset MLP. Training data were reduced through coreset selection to decrease computational requirements.

Pruned MLP. Low-importance network weights were removed to simplify the architecture.

4.3 Proposed Improvement: PCA-KNN

For the final stage of the project, a PCA-KNN pipeline was introduced. The workflow consisted of:

$$\text{StandardScaler} \rightarrow \text{PCA} \rightarrow \text{KNN}$$

The rationale behind this approach was that KNN depends heavily on distance calculations, making scaling and dimensionality reduction essential.

5 Experimental Setup

Experiments were conducted using Python with the following libraries: Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.

Evaluation metrics included:

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- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

A train-test split approach was used throughout the experiments.

6 Results

6.1 Auto MPG Dataset Results

Table 3 reports the performance of all models on the Auto MPG dataset.

Random Forest achieved the best performance ($R^2 = 0.913$, MSE $= 4.673$). MLP also performed strongly. PCA slightly reduced performance due to the already low dimensionality of the dataset. Pruning produced negligible reductions in predictive accuracy.

Table 3: Model Performance on the Auto MPG Dataset

Model	MSE	R^2
Linear Regression	8.196	0.848
SVR	6.814	0.873
MLP	5.122	0.905
Random Forest	**4.673**	**0.913**
PCA + MLP	6.150	0.886
Coreset MLP	7.268	0.865
Pruned MLP	5.166	0.904

6.2 Vehicles Dataset Results

Table 4 reports model performance on the Vehicles Fuel Efficiency dataset.

Table 4: Model Performance on the Vehicles Fuel Efficiency Dataset

Model	MSE	R^2
Linear Regression	97.963	0.200
SVR	121.856	0.005
MLP	116.083	0.052
Random Forest	**15.111**	**0.877**
PCA + MLP	16.786	0.863
Coreset MLP	108.010	0.118
Pruned MLP	118.766	0.031

Random Forest again demonstrated the highest predictive performance. The original MLP struggled substantially on the high-dimensional dataset. PCA dramatically improved MLP performance, increasing R^2 from 0.052 to 0.863, a gain of more than $16\times$.

6.3 Improvements and Optimization Results

To improve predictive performance while reducing computational complexity, several optimization techniques were explored throughout the project. These techniques were evaluated on both the Auto MPG dataset and the Vehicles Fuel Efficiency dataset. The results demonstrate that the effectiveness of a particular optimization strategy depends heavily on the characteristics of the dataset.

6.3.1 Improvements on the Auto MPG Dataset

The Auto MPG dataset is relatively small and low-dimensional. Consequently, the baseline models already achieved strong predictive performance. Compression techniques therefore focused on reducing complexity while preserving accuracy.

Table 5: Optimization Results on the Auto MPG Dataset

Model	MSE	R^2
Standard MLP	5.122	0.905
PCA + MLP	6.150	0.886
Coreset MLP	7.268	0.865
Pruned MLP	5.166	0.904
Random Forest	**4.673**	**0.913**
K-Means Coreset MLP	6.732	0.875

Several important observations can be made from Table 5:

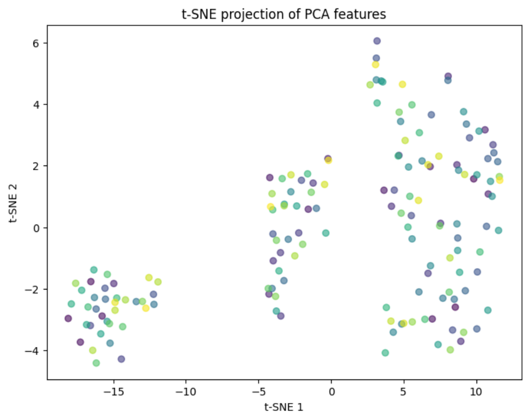

Figure 3:

- **Pruning was highly effective.** The Pruned MLP achieved an R^2 score of 0.904, almost identical to the Standard MLP (0.905). This suggests that many neural network weights were redundant and could be removed without significantly affecting predictive accuracy.
- **PCA slightly reduced performance.** The R^2 score decreased from 0.905 to 0.886. Since the Auto MPG dataset contains only a small number of informative features, dimensionality reduction removed useful predictive information.
- **Random coreset selection degraded performance.** Reducing the training set caused the R^2 score to decrease to 0.865, indicating that the smaller dataset did not sufficiently represent the original data distribution.
- **K-Means Coreset Selection improved upon random coreset selection.** By selecting representative samples from clusters rather than randomly, the MSE improved from 7.268 to 6.732 and the R^2 score increased from 0.865 to 0.875. Although it did not surpass the full MLP, it demonstrated that more intelligent coreset construction preserves information more effectively.
- **Random Forest remained the strongest model**, achieving the highest R^2 score of 0.913.

6.3.2 Improvements on the Vehicles Dataset

Unlike Auto MPG, the Vehicles dataset contained 47,702 observations and 84 features. The dataset included many categorical variables, redundant attributes, and substantial missing values. As a result, optimization techniques behaved very differently.

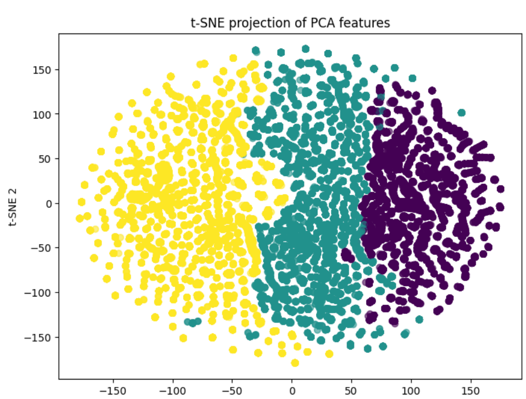

Figure 4:

The Vehicles dataset revealed several interesting findings from Table 6:

Table 6: Optimization Results on the Vehicles Fuel Efficiency Dataset

Model	MSE	R^2
Standard MLP	116.083	0.052
PCA + MLP	16.786	0.863
Coreset MLP	108.010	0.118
Pruned MLP	118.766	0.031
PCA + KNN	18.481	0.849
Random Forest	**15.111**	**0.877**

- **PCA dramatically improved neural network performance.** The Standard MLP achieved an R^2 score of only 0.052. After applying PCA, the R^2 score increased to 0.863. This substantial improvement suggests that the original feature space contained considerable redundancy and noise, which hindered neural network learning.
- **Pruning was ineffective.** Unlike the Auto MPG dataset, pruning reduced performance substantially. This indicates that larger, more complex datasets require richer representations and cannot tolerate aggressive weight removal.
- **Random coreset selection again performed poorly.** The reduction in training samples resulted in an R^2 score of only 0.118, highlighting the importance of maintaining sufficient diversity in large datasets.
- **The proposed PCA-KNN pipeline achieved strong performance.** By combining StandardScaler, PCA, and KNN, the model achieved an R^2 score of 0.849. Although Random Forest remained superior, PCA-KNN substantially outperformed the Standard MLP and provided a simpler alternative to neural network approaches.
- **Random Forest again emerged as the best-performing model**, achieving the highest R^2 score of 0.877.

7 Discussion

The findings demonstrate that dataset characteristics strongly influence model performance.

The Auto MPG dataset represented a relatively simple regression problem. Most models achieved satisfactory performance, with Random Forest producing the strongest results. Compression techniques such as pruning had minimal impact because the dataset contained only a small number of informative features.

The Vehicles dataset presented a more challenging task due to its size and dimensionality. Traditional neural networks struggled to learn meaningful representations from the raw feature space. However, PCA dramatically improved neural network performance by removing redundant and noisy features.

Compression techniques behaved differently across datasets. While pruning and coreset selection proved effective for simpler data, they reduced performance considerably on the Vehicles dataset.

The proposed PCA-KNN pipeline demonstrated that combining dimensionality reduction with distance-based learning provides a competitive alternative to more complex neural network approaches.

8 Comparison with Existing Studies

The findings align with previous literature. Aliyu et al. demonstrated the effectiveness of MLP models using the Auto MPG dataset; similarly, MLP-based approaches performed strongly on the Auto MPG dataset in this study. Sarkar and Hasan reported strong performance from ensemble methods, which this study confirms, with Random Forest consistently achieving the highest predictive accuracy. Çeken and Gemci highlighted the value of advanced machine learning methods for modern vehicle datasets; the present work extends these findings by incorporating compression techniques and dimensionality reduction strategies. Unlike previous studies, this work explicitly investigated the trade-off between model complexity and predictive performance.

9 Limitations

Several limitations should be acknowledged:
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- Single train-test splits were used instead of repeated cross-validation.
- Statistical significance tests and error bars were not computed.
- The study relied on static datasets rather than real-time vehicle data.
- Explainability techniques were not explored.

Future work should address these limitations through more robust evaluation procedures.

10 Future Work

Future studies may consider:
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- k-fold cross-validation.
- Friedman and Nemenyi statistical tests.
- Explainable AI (XAI) techniques.
- Real-time fuel efficiency prediction systems.
- Additional optimisation strategies.
- Hybrid ensemble approaches.

11 Conclusion

This study compared machine learning, neural network, and optimisation approaches for vehicle fuel efficiency prediction using both historical and modern datasets. Random Forest consistently achieved the highest predictive accuracy across both datasets. PCA proved highly effective in improving neural network performance on high-dimensional vehicle data, while pruning reduced complexity with minimal impact on simpler datasets. The proposed PCA-KNN pipeline demonstrated that dimensionality reduction combined with instance-based learning can provide a competitive and computationally efficient alternative to neural network approaches.

Overall, the results highlight the importance of selecting modelling techniques that align with dataset characteristics, and demonstrate that compression and optimisation methods can play a valuable role in practical machine learning applications for transportation and sustainability.

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